

Campus Event Board: Discovering and Managing Campus Events

What the Software Does

Campus events are announced across flyers, group chats, club email lists, and social media, so students miss events and organizers struggle to reach an audience. Campus Event Board is a web application that gives the campus community one place to discover, post, and RSVP to events. Visitors can browse upcoming events without an account. Once logged in, students can search by keyword and filter by category, date, and location, open a detail page for any event, and RSVP or cancel with one click. Events with a capacity limit show as "Full" when they fill up, and students can download any event as an .ics file to add to their personal calendar. Organizers can post and manage their own events through a form, view the list of attendees, and bulk-import a semester's schedule from a CSV file. Administrators can connect a public ICS calendar feed, which the system imports daily so existing university calendars appear automatically without duplicates. Administrators can also remove inappropriate listings, deactivate accounts, and manage categories.

Users

Students are the primary audience. They browse, search, and RSVP to events and keep a personal list of the events they plan to attend. Event organizers (student clubs, departments, and staff) create, edit, and cancel events, import them in bulk, and see who is attending. Administrators keep the platform healthy by moderating content, managing users and categories, and configuring external calendar feeds.

Technology and Rationale

User interface (React, HTML5, CSS). The front end is a responsive single-page application built with React. Reusable components (event cards, filter bar, RSVP button) keep the code organized, and the page updates instantly when a student RSVPs or changes a filter, without a full reload. Responsive CSS ensures the interface works on the phones students use most, and semantic HTML supports keyboard and screen reader accessibility.

Server (Node.js and Express). The back end is a REST API written in JavaScript, so the same language is used across the whole project, which speeds up development and simplifies debugging. Express handles routing, and middleware enforces role-based permissions on every request. Supporting libraries include bcrypt for password hashing, JSON Web Tokens for login sessions, csv-parse for imports, node-ical for reading calendar feeds, and node-cron to schedule the daily sync.

Database (PostgreSQL). The data is structured and highly relational: organizers own events, events belong to categories, and students connect to events through RSVPs. A relational database enforces these relationships with foreign keys and prevents duplicate RSVPs with a unique constraint. PostgreSQL's transactions and row locking guarantee that two students clicking RSVP at the same moment can never push an event past its capacity, and its full-text search and date indexing keep keyword and date-range queries fast. The Express server connects to PostgreSQL through the node-postgres (pg) library using a connection pool, with parameterized queries to prevent SQL injection.

Tooling and deployment. The code lives in a GitHub repository, with user stories tracked as Issues, and automated tests (Jest) cover critical logic such as capacity limits and CSV validation. The application will be deployed with HTTPS on a cloud platform such as Render, which also hosts the managed PostgreSQL instance.